

CS-600: Software Design and Development

2-1 Short Paper: Visualizing Business Strategy for Application Development

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Importance of Connection Between Business Strategy and Software Engineering

As software engineers, we need to understand how businesses run to develop applications that meet real organizational needs. It is important to recognize how systems support companies and contribute to their success. For this reason, connecting business strategy to software engineering is essential. One effective way to achieve this is by using tools such as the business model canvas and Wardley maps. In this paper, I examined the Alpha Pet case and its business model. I then explain how the business model canvas informs application development, followed by an extended analysis using a Wardley map. Finally, I reflect on how this process relates to my future professional practice.

Business Model Canvas for Alpha Pet

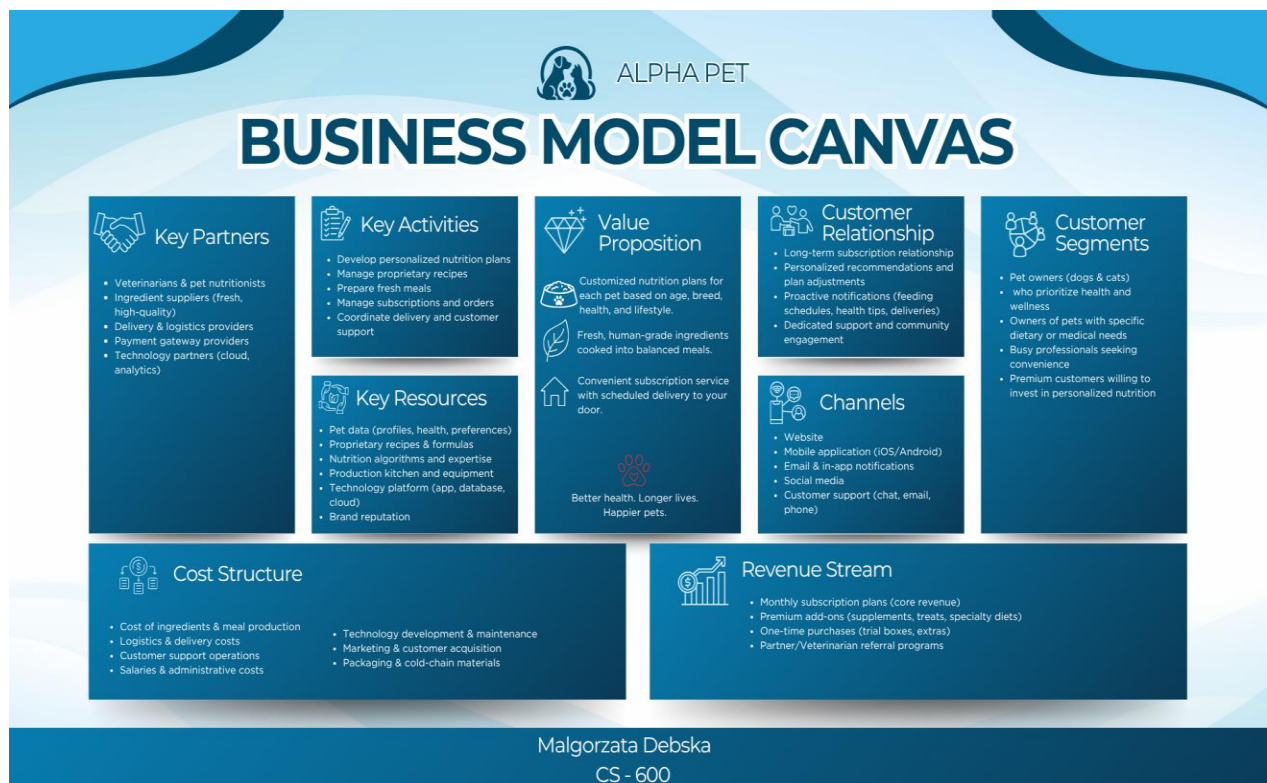
Alpha Pet is a specialty pet store that provides customized meal plans for dogs and cats based on their individual needs. Instead of selling standard pet food, the company develops personalized nutrition plans that consider factors such as age, breed, weight, and health conditions. Because of this approach, the system supporting the business must include features for managing pet data, feeding schedules, and proprietary recipes. The company's customer segment consists of pet owners who want to improve their pet's health and are willing to invest in personalized nutrition. Its value proposition focuses on delivering high-quality, tailored meals that are more beneficial than traditional pet food.

Customer interactions take place primarily through digital platforms such as websites and mobile applications. These platforms allow users to input pet information, receive customized plans, and

manage subscriptions. Since Alpha Pet uses a subscription model, customer relationships are ongoing and long-term.

Revenue is generated mainly through subscription payments. Key business activities include developing nutrition plans, preparing meals, managing recipes, and coordinating deliveries.

Important resources include pet data, nutritional formulas, and proprietary recipes. The company also relies on partnerships with suppliers, veterinarians, and delivery services to support quality and efficiency. The business model canvas offers a structured view of how Alpha Pet uses highlights the role of technology in supporting its processes.



How the Canvas Informs Application Development

The business model canvas is not only useful for business planning but also plays a significant role in software design. It helps developers better understand business needs and translate them into system features. One major requirement is the ability to store and manage detailed pet profiles. This calls for a secure and scalable database that includes information such as age, breed, weight, and medical history. Another important feature is scheduling. The system should allow users to view feeding plans and receive reminders for feeding times and deliveries, improving both convenience and consistency. Recipe management is also essential, as the company relies on proprietary formulas. The system must securely store and manage these recipes while allowing updates when needed. In addition, the application must support core e-commerce functions such as subscription management, payment processing, and order tracking. Customers should be able to manage their plans easily and monitor deliveries in real time. Delivery coordination is important. The system should integrate with inventory management and delivery services to ensure correct and prompt fulfillment. By using the canvas, developers can clearly connect business requirements to technical solutions, ensuring that the system aligns with organizational goals.

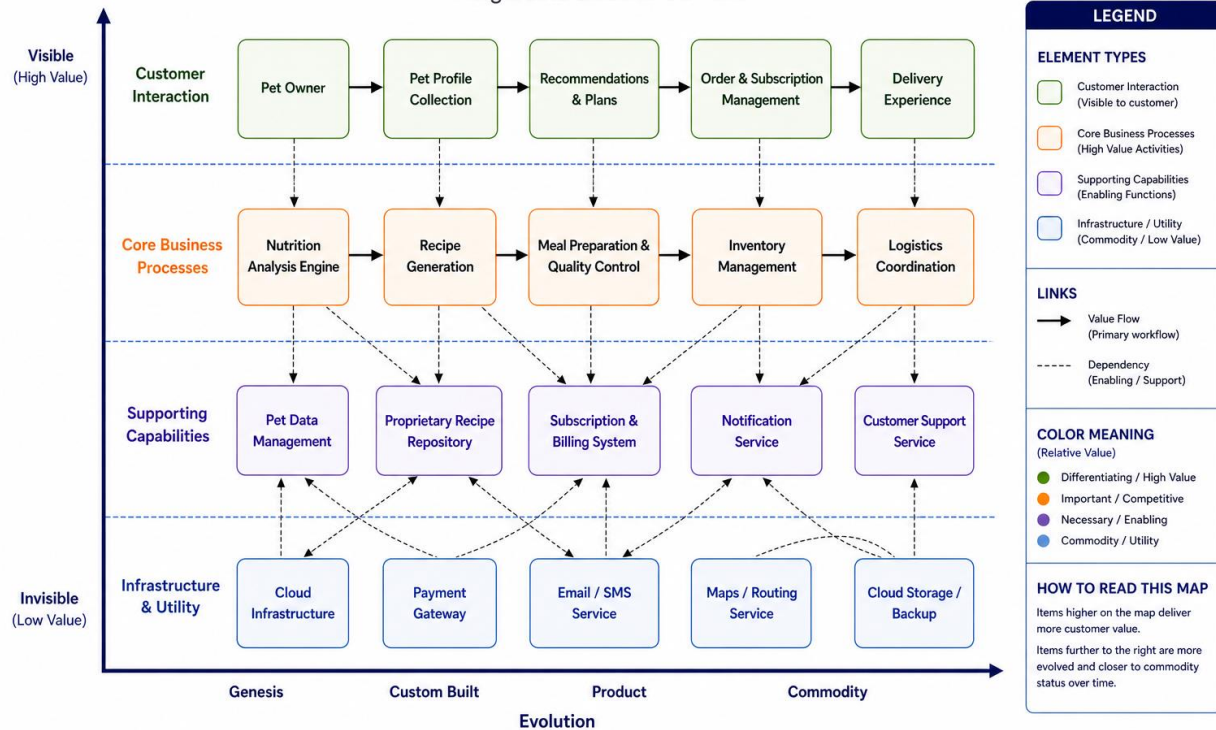
Wardley Map and Domain Workflows

While the business model canvas provides a strategic overview, the Wardley map offers a deeper look at system workflows and dependencies. It helps visualize how different components interact and evolve over time. For Alpha Pet, the primary users are pet owners looking for convenient and reliable ways to provide healthy food for their pets. Their main needs include personalized meal plans, easy ordering, and dependable delivery. The workflow begins with data collection, where users enter pet information into the system. This data is then analyzed to figure out proper nutrition plans. Based on this analysis, recipes are generated and used to prepare meals. The process continues with order placement and subscription management, followed by delivery to the customer.

Domain Workflow Visualization

ALPHA PET – WRADLEY MAP

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This sequence illustrates how each part is connected and dependent on the others.

Insights from Wardley Maps: Dependencies, Value Chains, and Innovation Opportunities

Wardley maps are useful for finding system dependencies. In Alpha Pet's case, the entire process depends on exact and complete pet data. Poor data quality can negatively affect recommendations, while delivery performance relies on effective inventory and logistics systems. The map also highlights the value chain. Personalized nutrition planning is a high-value, differentiating feature of the business. In contrast, elements like payment processing and cloud services are more standardized and can be treated as commodities. Another advantage is the ability to find opportunities for innovation. Alpha Pet could enhance its services by incorporating artificial intelligence to improve recommendations based on historical data. Real-time delivery tracking could further improve customer experience. Additionally, offering health

analytics for pets could provide extra value and strengthen customer engagement. These insights help guide strategic decisions and prioritize investments.

Connection to Future Professional Practice

Understanding how to connect business strategy with software design is highly relevant for a career in software engineering. Building effective systems requires more than technical skills; it also involves understanding users, business goals, and real-world problems. Tools like the business model canvas and Wardley maps can improve communication between developers, stakeholders, and business teams. They make it easier to name priorities and design systems that deliver meaningful results. Applying these techniques in future work will help ensure that the systems I develop are both functional and aligned with business goals

Conclusion

The Alpha Pet case proves that business strategy can be effectively linked to software design. The business model canvas provides a clear understanding of operations, while the Wardley map offers insight into workflows, dependencies, and innovation opportunities. Together, these tools support the development of systems that are practical, efficient, and aligned with business goals.

References

Chandrasekaran, P., & Krishnan, K. (2022). *Domain-driven design with Java: A practitioner's guide*. Packt Publishing.

